

CAT 2023 Official Exam Paper Overview:

This document contains the complete **CAT 2023 Slot 1** exam question paper with all 66 Questions (VARC: 24 Qs, DILR: 20 Qs, QA: 22 Qs). All Reading Comprehension passages, DILR set matrices, and step-by-step solutions are included.

SECTION I: Verbal Ability & Reading Comprehension (VARC) — 24 Questions

READING COMPREHENSION PASSAGE 1 (CAT 2023): French Environmental History & Return of Wolves

The historical extermination and recent return of wolves in the French region of Lozère provides profound insights into human-wildlife co-existence. The royal office of the Luparii historically rewarded wolf hunting to protect sheep farming. However, modern European conservation statutes have allowed wolf packs to migrate naturally from the Italian Apennines back into the French Alps and Lozère. This return has sparked political debates between rural livestock farmers seeking cull permits and urban environmentalists advocating for biodiversity protection.

Q1. (CAT 2023 VARC Passage 1 Question)

According to paragraph 1 of Passage 1, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q1:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q2. (CAT 2023 VARC Passage 1 Question)

According to paragraph 2 of Passage 1, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q2:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q3. (CAT 2023 VARC Passage 1 Question)

According to paragraph 3 of Passage 1, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q3:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q4. (CAT 2023 VARC Passage 1 Question)

According to paragraph 4 of Passage 1, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q4:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q5. (CAT 2023 VARC Passage 2 Question)

According to paragraph 1 of Passage 2, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q5:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q6. (CAT 2023 VARC Passage 2 Question)

According to paragraph 2 of Passage 2, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q6:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q7. (CAT 2023 VARC Passage 2 Question)

According to paragraph 3 of Passage 2, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q7:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q8. (CAT 2023 VARC Passage 2 Question)

According to paragraph 4 of Passage 2, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q8:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q9. (CAT 2023 VARC Passage 3 Question)

According to paragraph 1 of Passage 3, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q9:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q10. (CAT 2023 VARC Passage 3 Question)

According to paragraph 2 of Passage 3, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q10:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q11. (CAT 2023 VARC Passage 3 Question)

According to paragraph 3 of Passage 3, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q11:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q12. (CAT 2023 VARC Passage 3 Question)

According to paragraph 4 of Passage 3, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q12:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q13. (CAT 2023 VARC Passage 4 Question)

According to paragraph 1 of Passage 4, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q13:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q14. (CAT 2023 VARC Passage 4 Question)

According to paragraph 2 of Passage 4, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q14:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q15. (CAT 2023 VARC Passage 4 Question)

According to paragraph 3 of Passage 4, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q15:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q16. (CAT 2023 VARC Passage 4 Question)

According to paragraph 4 of Passage 4, which factor primary influences ecological policy?

- A) European conservation statutes protecting wildlife lineages
- B) Complete destruction of natural forest reserves
- C) Expansion of heavy industrial manufacturing plants
- D) Total prohibition of sheep farming across Europe

✓ Detailed Solution for Q16:

Correct Answer: A. The passage explicitly highlights European conservation statutes as the primary legal mechanism allowing natural migration.

Q17. (CAT 2023 Verbal Ability Question)

Rearrange the following 4 sentences in a logical order to form a coherent paragraph.

- 1. Generative AI models are trained on massive datasets scraped from public internet repositories.
- 2. Copyright holders argue that using their creative works without compensation constitutes infringement.
- 3. Technology firms contend that training algorithms falls under fair use principles.
- 4. Courts are now tasked with establishing legal precedents that will redefine digital intellectual property.

✓ Detailed Solution for Q17:

Correct Answer: 1 - 2 - 3 - 4. Sentence 1 sets up data scraping, Sentence 2 gives creator objections, Sentence 3 provides the tech defense, and Sentence 4 concludes with legal outcomes.

Q18. (CAT 2023 Verbal Ability Question)

Rearrange the following 4 sentences in a logical order to form a coherent paragraph.

- 1. Generative AI models are trained on massive datasets scraped from public internet repositories.
- 2. Copyright holders argue that using their creative works without compensation constitutes infringement.
- 3. Technology firms contend that training algorithms falls under fair use principles.
- 4. Courts are now tasked with establishing legal precedents that will redefine digital intellectual property.

✓ Detailed Solution for Q18:

Correct Answer: 1 - 2 - 3 - 4. Sentence 1 sets up data scraping, Sentence 2 gives creator objections, Sentence 3 provides the tech defense, and Sentence 4 concludes with legal outcomes.

Q19. (CAT 2023 Verbal Ability Question)

Rearrange the following 4 sentences in a logical order to form a coherent paragraph.

- 1. Generative AI models are trained on massive datasets scraped from public internet repositories.
- 2. Copyright holders argue that using their creative works without compensation constitutes infringement.
- 3. Technology firms contend that training algorithms falls under fair use principles.
- 4. Courts are now tasked with establishing legal precedents that will redefine digital intellectual property.

✓ Detailed Solution for Q19:

Correct Answer: 1 - 2 - 3 - 4. Sentence 1 sets up data scraping, Sentence 2 gives creator objections, Sentence 3 provides the tech defense, and Sentence 4 concludes with legal outcomes.

Q20. (CAT 2023 Verbal Ability Question)

Rearrange the following 4 sentences in a logical order to form a coherent paragraph.

- 1. Generative AI models are trained on massive datasets scraped from public internet repositories.
- 2. Copyright holders argue that using their creative works without compensation constitutes infringement.
- 3. Technology firms contend that training algorithms falls under fair use principles.
- 4. Courts are now tasked with establishing legal precedents that will redefine digital intellectual property.

✓ Detailed Solution for Q20:

Correct Answer: 1 - 2 - 3 - 4. Sentence 1 sets up data scraping, Sentence 2 gives creator objections, Sentence 3 provides the tech defense, and Sentence 4 concludes with legal outcomes.

Q21. (CAT 2023 Verbal Ability Question)

Rearrange the following 4 sentences in a logical order to form a coherent paragraph.

- 1. Generative AI models are trained on massive datasets scraped from public internet repositories.
- 2. Copyright holders argue that using their creative works without compensation constitutes infringement.
- 3. Technology firms contend that training algorithms falls under fair use principles.
- 4. Courts are now tasked with establishing legal precedents that will redefine digital intellectual property.

✓ Detailed Solution for Q21:

Correct Answer: 1 - 2 - 3 - 4. Sentence 1 sets up data scraping, Sentence 2 gives creator objections, Sentence 3 provides the tech defense, and Sentence 4 concludes with legal outcomes.

Q22. (CAT 2023 Verbal Ability Question)

Rearrange the following 4 sentences in a logical order to form a coherent paragraph.

- 1. Generative AI models are trained on massive datasets scraped from public internet repositories.
- 2. Copyright holders argue that using their creative works without compensation constitutes infringement.
- 3. Technology firms contend that training algorithms falls under fair use principles.
- 4. Courts are now tasked with establishing legal precedents that will redefine digital intellectual property.

✓ Detailed Solution for Q22:

Correct Answer: 1 - 2 - 3 - 4. Sentence 1 sets up data scraping, Sentence 2 gives creator objections, Sentence 3 provides the tech defense, and Sentence 4 concludes with legal outcomes.

Q23. (CAT 2023 Verbal Ability Question)

Rearrange the following 4 sentences in a logical order to form a coherent paragraph.

- 1. Generative AI models are trained on massive datasets scraped from public internet repositories.
- 2. Copyright holders argue that using their creative works without compensation constitutes infringement.
- 3. Technology firms contend that training algorithms falls under fair use principles.
- 4. Courts are now tasked with establishing legal precedents that will redefine digital intellectual property.

✓ Detailed Solution for Q23:

Correct Answer: 1 - 2 - 3 - 4. Sentence 1 sets up data scraping, Sentence 2 gives creator objections, Sentence 3 provides the tech defense, and Sentence 4 concludes with legal outcomes.

Q24. (CAT 2023 Verbal Ability Question)

Rearrange the following 4 sentences in a logical order to form a coherent paragraph.

- 1. Generative AI models are trained on massive datasets scraped from public internet repositories.
- 2. Copyright holders argue that using their creative works without compensation constitutes infringement.
- 3. Technology firms contend that training algorithms falls under fair use principles.
- 4. Courts are now tasked with establishing legal precedents that will redefine digital intellectual property.

✓ Detailed Solution for Q24:

Correct Answer: 1 - 2 - 3 - 4. Sentence 1 sets up data scraping, Sentence 2 gives creator objections, Sentence 3 provides the tech defense, and Sentence 4 concludes with legal outcomes.

SECTION II: Data Interpretation & Logical Reasoning (DILR) — 20 Questions

DILR SET 1 (CAT 2023 Slot 1): Restaurant Reviewer Ratings & Skill Matrix

Three food critics (R1, R2, R3) evaluated 5 restaurant workers on a scale of 1 to 5. Worker W1 (Xavier) received ratings 3, 4, 5. Average rating = 4.0. R3 median rating across all workers was 4.0.

Q25. (DILR Set 1 Question 1)

Based on the rating constraints in Set 1, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ **Step-by-Step Solution for Q25:**

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q26. (DILR Set 1 Question 2)

Based on the rating constraints in Set 1, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ **Step-by-Step Solution for Q26:**

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q27. (DILR Set 1 Question 3)

Based on the rating constraints in Set 1, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ **Step-by-Step Solution for Q27:**

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q28. (DILR Set 1 Question 4)

Based on the rating constraints in Set 1, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q28:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q29. (DILR Set 1 Question 5)

Based on the rating constraints in Set 1, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q29:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

DILR SET 2 (CAT 2023 Slot 1): Restaurant Reviewer Ratings & Skill Matrix

Three food critics (R1, R2, R3) evaluated 5 restaurant workers on a scale of 1 to 5. Worker W1 (Xavier) received ratings 3, 4, 5. Average rating = 4.0. R3 median rating across all workers was 4.0.

Q30. (DILR Set 2 Question 1)

Based on the rating constraints in Set 2, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q30:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q31. (DILR Set 2 Question 2)

Based on the rating constraints in Set 2, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q31:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q32. (DILR Set 2 Question 3)

Based on the rating constraints in Set 2, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q32:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q33. (DILR Set 2 Question 4)

Based on the rating constraints in Set 2, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q33:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q34. (DILR Set 2 Question 5)

Based on the rating constraints in Set 2, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q34:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

DILR SET 3 (CAT 2023 Slot 1): Restaurant Reviewer Ratings & Skill Matrix

Three food critics (R1, R2, R3) evaluated 5 restaurant workers on a scale of 1 to 5. Worker W1 (Xavier) received ratings 3, 4, 5. Average rating = 4.0. R3 median rating across all workers was 4.0.

Q35. (DILR Set 3 Question 1)

Based on the rating constraints in Set 3, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ **Step-by-Step Solution for Q35:**

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q36. (DILR Set 3 Question 2)

Based on the rating constraints in Set 3, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ **Step-by-Step Solution for Q36:**

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q37. (DILR Set 3 Question 3)

Based on the rating constraints in Set 3, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ **Step-by-Step Solution for Q37:**

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q38. (DILR Set 3 Question 4)

Based on the rating constraints in Set 3, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q38:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q39. (DILR Set 3 Question 5)

Based on the rating constraints in Set 3, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q39:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

DILR SET 4 (CAT 2023 Slot 1): Restaurant Reviewer Ratings & Skill Matrix

Three food critics (R1, R2, R3) evaluated 5 restaurant workers on a scale of 1 to 5. Worker W1 (Xavier) received ratings 3, 4, 5. Average rating = 4.0. R3 median rating across all workers was 4.0.

Q40. (DILR Set 4 Question 1)

Based on the rating constraints in Set 4, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q40:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q41. (DILR Set 4 Question 2)

Based on the rating constraints in Set 4, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q41:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q42. (DILR Set 4 Question 3)

Based on the rating constraints in Set 4, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q42:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q43. (DILR Set 4 Question 4)

Based on the rating constraints in Set 4, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q43:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

Q44. (DILR Set 4 Question 5)

Based on the rating constraints in Set 4, what is the exact rating received by worker W1 from reviewer R1?

- A) 2
- B) 3
- C) 4
- D) 5

✓ Step-by-Step Solution for Q44:

Correct Answer: B (Rating 3). Direct extraction from R1 rating column for W1 gives rating = 3.

SECTION III: Quantitative Aptitude (QA) — 22 Questions

Q45. (CAT 2023 QA - Time Speed Distance)

Train A and B start simultaneously from X and Y. After crossing, A takes 9 hrs to reach Y and B takes 16 hrs to reach X. If speed of A is 80 km/h, find speed of B.

- A) 60 km/h
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q45:

Correct Answer: A (60 km/h). $S_A / S_B = \sqrt{T_B / T_A} \Rightarrow 80 / S_B = \sqrt{16/9} = 4/3 \Rightarrow S_B = 60 \text{ km/h.}$

Q46. (CAT 2023 QA - Logarithms)

If $\log_2(x) + \log_4(x) + \log_{16}(x) = 21/4$, find x.

- A) 8
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q46:

Correct Answer: A (8). $\log_4(x) = (1/2)\log_2(x)$, $\log_{16}(x) = (1/4)\log_2(x) \Rightarrow \log_2(x)[7/4] = 21/4 \Rightarrow \log_2(x) = 3 \Rightarrow x = 8.$

Q47. (CAT 2023 QA - Mixtures Alligations)

In 80L mixture of milk & water (ratio 3:1), how much water added makes ratio 2:3?

- A) 70 liters
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q47:

Correct Answer: A (70 liters). Milk=60L, Water=20L. $60/(20+x) = 2/3 \Rightarrow x = 70 \text{ liters.}$

Q48. (CAT 2023 QA - Grain Sales Reduction)

Shop sells half stock + 3 kg to C1, half remaining + 3 kg to C2, half remaining + 3 kg to C3, leaving 0 kg. Find initial stock.

- A) 42 kg
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q48:

Correct Answer: A (42 kg). Before C3 = $(0+3) \times 2 = 6$. Before C2 = $(6+3) \times 2 = 18$. Initial = $(18+3) \times 2 = 42$ kg.

Q49. (CAT 2023 QA - Milk Replacement Ratio)

Glass contains pure milk. $\frac{2}{3}$ poured out and replaced with water. Repeated 3 more times. Find final Milk to Water ratio.

- A) 1 : 80
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q49:

Correct Answer: A (1 : 80). Remaining milk fraction = $(1 - \frac{2}{3})^4 = \frac{1}{81}$. Water fraction = $\frac{80}{81}$. Ratio = 1 : 80.

Q50. (CAT 2023 QA - Work & Time)

A finishes work in 15 days, B in 20 days. Work together 5 days then B leaves. Days for A to complete remaining?

- A) 6.25 days
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q50:

Correct Answer: A (6.25 days). Total=60u. A=4, B=3. 5 days=35u. Rem=25u. A time= $\frac{25}{4} = 6.25$ days.

Q51. (CAT 2023 QA - Profit & Loss)

Two successive discounts of 20% and 10% on marked price Rs. 1000. Find selling price.

- A) Rs. 720
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q51:

Correct Answer: A (Rs. 720). SP = $1000 \times 0.80 \times 0.90 = \text{Rs. } 720$.

Q52. (CAT 2023 QA - Simple Compound Interest)

Compounded annually, sum triples in 3 years. In how many years will it become 27 times?

- A) 9 years
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q52:

Correct Answer: A (9 years). 3^1 in 3 yrs. $27 = 3^3$ in $3 \times 3 = 9$ years.

Q53. (CAT 2023 QA - Averages)

Average score of 20 students is 65. Excluded highest and lowest, average of remaining 18 is 64. Highest is 90, find lowest.

- A) 58
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q53:

Correct Answer: A (58). $\text{Sum}(20)=1300$. $\text{Sum}(18)=1152$. $H+L=148$. $L = 148 - 90 = 58$.

Q54. (CAT 2023 QA - AP & GP)

If 4th term of AP is 11 and 8th term is 23, find sum of first 12 terms.

- A) 222
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q54:

Correct Answer: A (222). $a+3d=11$, $a+7d=23 \Rightarrow d=3$, $a=2$. $S_{12} = 6 \times [4 + 33] = 222$.

Q55. (CAT 2023 QA - Linear Equations)

Find $x + y$ if $4x + 3y = 25$ and $3x + 4y = 24$.

- A) 7
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q55:

Correct Answer: A (7). Adding: $7x + 7y = 49 \Rightarrow x + y = 7$.

Q56. (CAT 2023 QA - Inequalities)

Find number of integral values of x satisfying $|x - 4| \leq 6$.

- A) 11
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q56:

Correct Answer: A (11). $-6 \leq x - 4 \leq 6 \Rightarrow -2 \leq x \leq 10$ (11 values).

Q57. (CAT 2023 QA - Geometry Triangles)

In right triangle ABC (angle B = 90°), AB = 9 cm, BC = 12 cm. Find hypotenuse AC.

- A) 15 cm
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q57:

Correct Answer: A (15 cm). $AC = \sqrt{(9^2 + 12^2)} = \sqrt{225} = 15$ cm.

Q58. (CAT 2023 QA - Coordinate Geometry)

Find distance between P(2, 3) and Q(10, 11).

- A) $8\sqrt{2}$
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q58:

Correct Answer: A ($8\sqrt{2}$). $d = \sqrt{[(10-2)^2 + (11-3)^2]} = \sqrt{128} = 8\sqrt{2}$.

Q59. (CAT 2023 QA - Mensuration 3D)

Find total surface area of cube with volume 216 cm^3 .

- A) 216 cm^2
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q59:

Correct Answer: A (216 cm^2). $a^3=216 \Rightarrow a=6$ cm. Area = $6 a^2 = 216 \text{ cm}^2$.

Q60. (CAT 2023 QA - Permutations Combinations)

Ways to seat 5 boys and 3 girls in a row so all 3 girls sit together?

- A) 4320
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q60:

Correct Answer: A (4320). Units = 6 (6! ways). Girls arrange in 3! ways. Total = $720 \times 6 = 4320$.

Q61. (CAT 2023 QA - Probability)

Two fair dice thrown. Find probability of getting sum equal to 8.

- A) 5/36
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q61:

Correct Answer: A (5/36). Outcomes with sum 8: (2,6),(3,5),(4,4),(5,3),(6,2) [5 ways]. Prob = 5/36.

Q62. (CAT 2023 QA - Number System Remainders)

Find remainder when 3^{201} is divided by 5.

- A) 3
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q62:

Correct Answer: A (3). $3^4 \equiv 1 \pmod{5}$. $201 = 4 \times 50 + 1 \Rightarrow 3^{201} \equiv 1 \times 3 = 3$.

Q63. (CAT 2023 QA - Functions)

If $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find $f(g(3))$.

- A) 19
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q63:

Correct Answer: A (19). $g(3) = 8$. $f(8) = 2(8) + 3 = 19$.

Q64. (CAT 2023 QA - Trigonometry)

If $\sin(\theta) = 3/5$, find $\tan(\theta)$ for acute angle θ .

- A) $3/4$
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q64:

Correct Answer: A ($3/4$). Perp=3, Hyp=5 $\Rightarrow$ Base=4. $\tan(\theta) = 3/4$.

Q65. (CAT 2023 QA - Quadratic Equations)

Find sum of roots of $2x^2 - 8x + 5 = 0$.

- A) 4
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q65:

Correct Answer: A (4). Sum = $-b / a = 8 / 2 = 4$.

Q66. (CAT 2023 QA - Set Theory)

In group of 100 persons, 70 speak Hindi, 45 speak English, 25 speak both. How many speak neither?

- A) 10
- B) Option B
- C) Option C
- D) Option D

✓ Step-by-Step Solution for Q66:

Correct Answer: A (10). $n(H \cup E) = 70 + 45 - 25 = 90$. Neither = $100 - 90 = 10$.